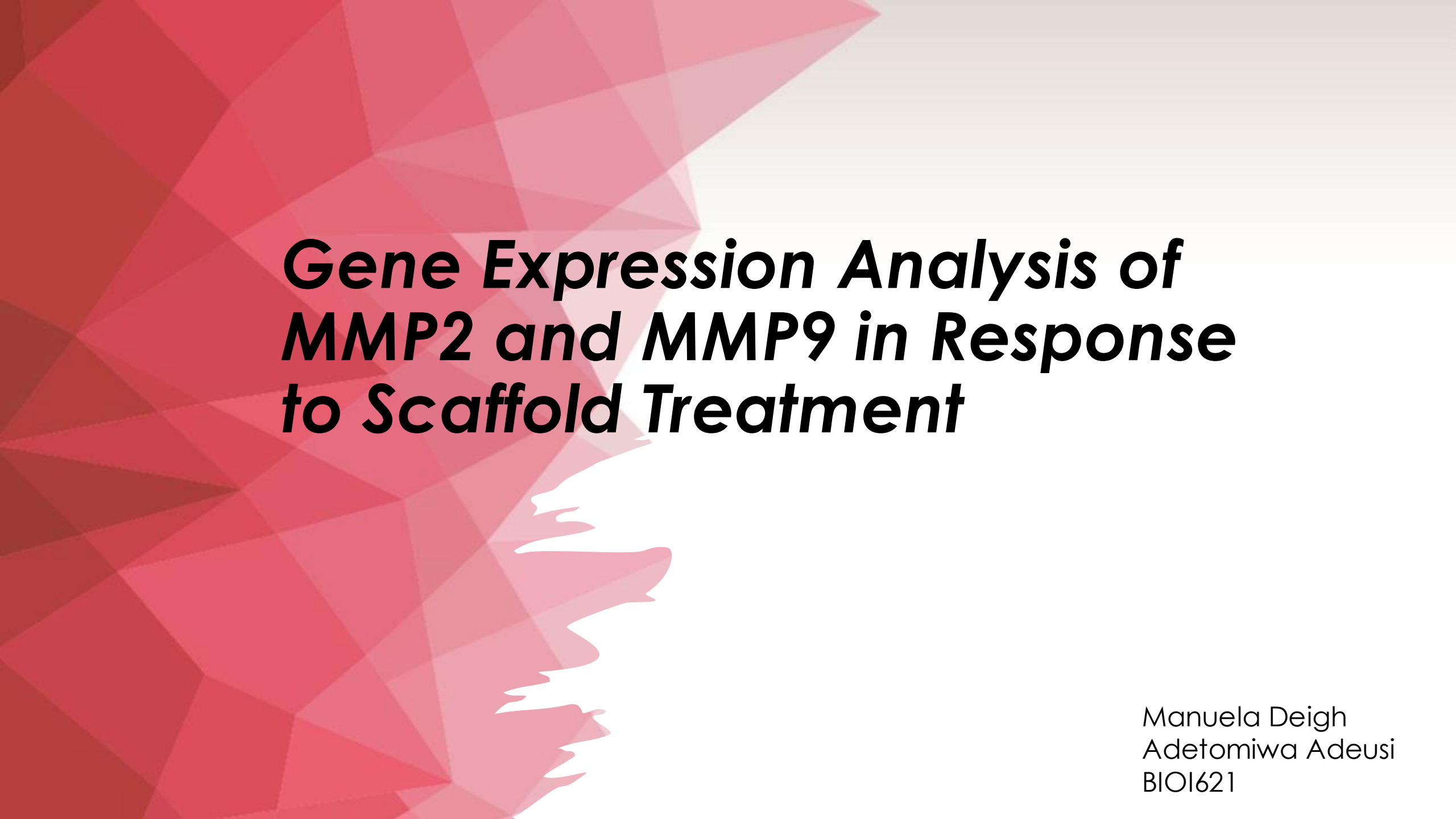




Gene Expression Analysis of MMP2 and MMP9 in Response to Scaffold Treatment

Manuela Deigh
Adetomiwa Adeusi
BIO1621

Research Question

How does macrophage MMP-2 and MMP-9 expression vary in response to zinc-based calcium phosphate (CaP) scaffolds with ion modifications?

- **Goal:** Our goal is to understand how macrophage response vary based on the design of biomaterials that promote tissue repair and immune regulation.

Background:

- Matrix Metalloproteinase-9 and Matrix Metalloproteinase-2 are enzymes that work by breaking down components of the extracellular matrix. In macrophages especially, they are important in:
 - a. Tissue remodelling
 - b. Inflammatory responses
 - c. Wound healing
 - d. Blood vessel formation
- Tissue engineering materials like calcium phosphate (CaP) scaffolds are used to promote bone regeneration and healing.
- Recent studies suggest **zinc-based ion modifications** (Zn, Mn, Li) to CaP scaffolds enhance immune modulation by influencing macrophage behavior.
- **Macrophages**, particularly M2-polarized types, are key players in tissue remodeling and osseointegration.

Data Overview

• Dataset

- GEO data: [GSE224524](#)
 - Organism: Mus musculus — RAW264.7 macrophage
 - Experimental Design:
 - Macrophages exposed to three different scaffold materials:
 - **CaP Zn** - zinc-based calcium phosphate scaffold
 - **CaP Zn0.8Mn** - zinc-based scaffold with 0.8% manganese
 - **CaP Zn0.8Mn0.1Li** - zinc-based scaffold with 0.8% manganese and 0.1% lithium
 - Goal of original study:
 - These scaffold materials were designed to investigate how zinc alloy scaffolds regulate macrophage polarization toward the **M2** (anti-inflammatory) phenotype

• Why we choose this data:

- Closely models macrophage response to biomaterials
- Uses the same macrophage cell line in Manuela's lab (RAW264.7)
- The original study focused on M2 polarization; our project focuses on **MMP-2** and **MMP-9** expression.
- Two replicates per material type → **6 samples total**.

Workflow Overview/ Tools

- **Download the Data**

- *Tools:*

- **fasterq-dump:** This tool is part of the SRA ToolKit. We used it to download the raw RNA-seq data directly from the NCBI Sequence Read Archive (SRA).

- **Quality Control (QC)**

- *Tools:*

- **FastQC :** We used to tool to check the quality of the downloaded dataset. This tool generated visual reports of our data, based on quality scores, GC content, sequence length distribution, and potential contamination like adapter content.
 - **MultiQC:** We also used this tool to generate aggregate QC reports of our multiple samples. The results were combined into a single interactive HTML report, and it helped us compare quality metrics of all our samples.

- **Read Alignment**

- *Tools:*

- **STAR:** We ran assembly using STAR as a backup option. This allowed comparative analysis, and the results were similar, but we ultimately moved on with the results from HISAT2.
 - **HISAT2 :** A fast and sensitive alignment program for mapping RNA-seq reads to a reference genome. HISAT2 uses a graph-based index to efficiently handle splicing events.
 - **SAMtools:** A toolkit for working with .sam and .bam files. It converts, sorts, and indexes alignment files for downstream analysis.

- **Transcriptome Assembly & Annotation:**

- *Tools*

- **StringTie:** We used to tool to assemble the transcripts from the aligned RNA-seq reads. We moved on with this one because It can estimate transcript abundances (e.g., FPKM values) and perform reference-guided assembly with an annotation file
 - This step also annotates the assembled transcripts by linking them to known gene features from the GENCODE reference.

- **Gene Expression Analysis & Visualization**

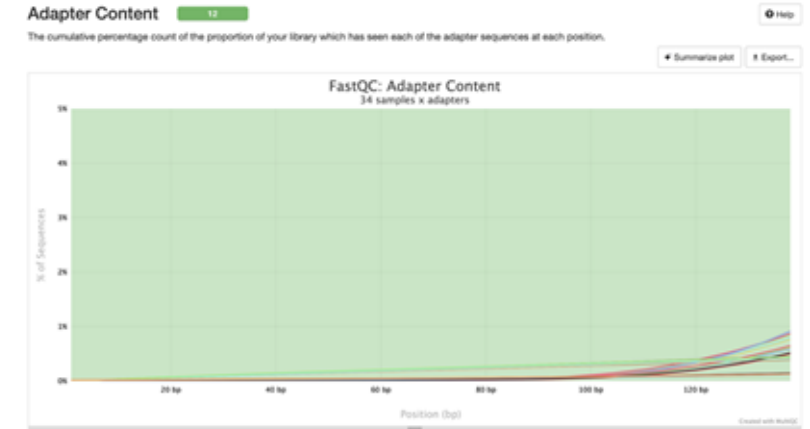
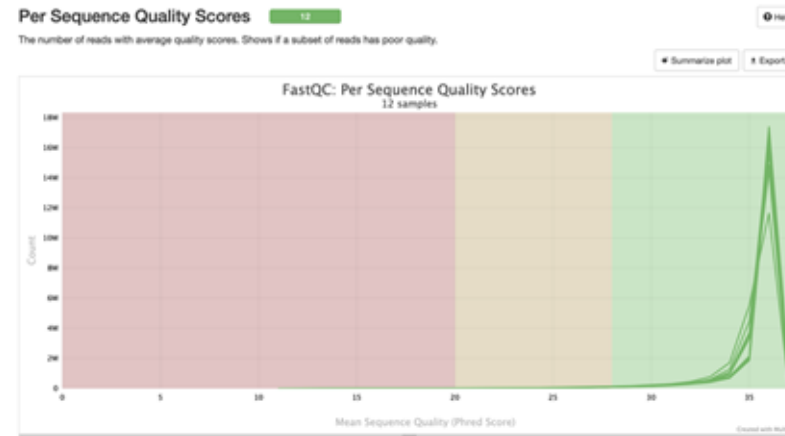
- *Tools:*

- **R (dplyr, ggplot2):** Used for data cleaning, filtering for genes of interest (Mmp2 and Mmp9), computing average FPKM by treatment, and visualizing expression patterns.

Data Download/ Quality Control

- **Dataset:**

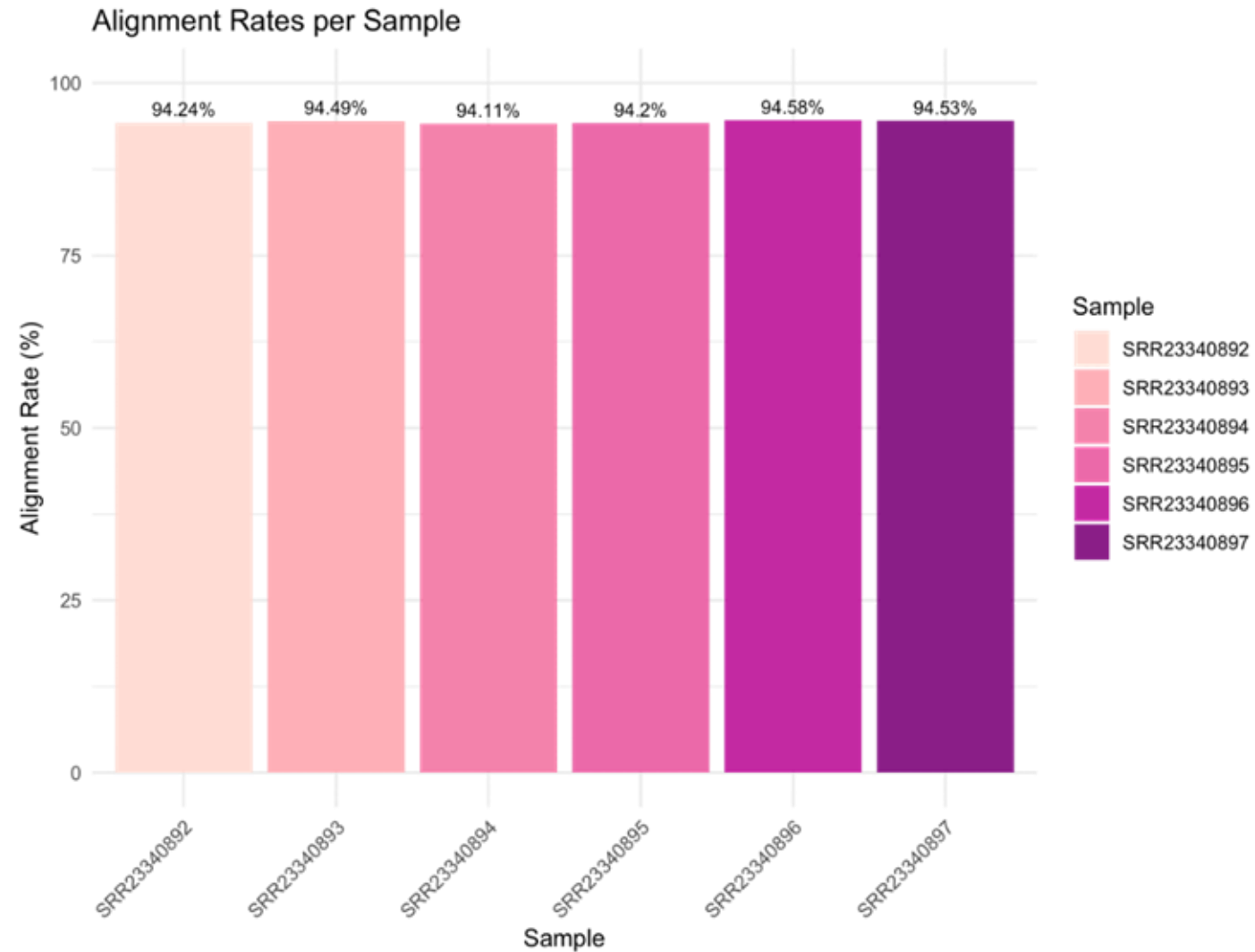
- We used fasterq-dump to download 6 paired-end RNA-seq samples (FASTQ format) of mouse macrophages exposed to different ion-modified calcium phosphate (CaP) scaffolds.
- Once downloaded, we ran FastQC on all 6 FASTQ files to assess quality
- Then we used MultiQC to summarize and visualize result across all samples



The outputs above show that all six RNA-seq samples had consistently high sequence quality. The Mean Phred scores stayed above 30 across all base positions, indicating low error rates. Most reads had an average Phred score around 35. Adapter contamination was minimal and did not require trimming. We then use those files for alignment

Read alignment

- We download a reference genome, which was the mm10 mouse genome index from the JHU HISAT2 website
- Created a custom script to:
 - Align paired-end FASTQ files to the reference genome using **HISAT2**
 - Convert SAM to BAM, then sort and index using **SAMtools**
 - Approximate runtime: **~1 hour per sample**
- **Output Files:**
 - SRRXXXXX_sorted.bam
 - SRRXXXXX_sorted.bam.bai
 - SRRXXXXX_hisat2_summary.txt
- The bar plot shows alignment rates for each sample. All samples achieved high alignment rates (~90%+), which indicating good-quality data and a well-matched reference genome.
- We then use those bam files for assembly and quantification which lead us to our next slides



Transcriptome Assembly & Annotation

- **Annotation:**

- Used GENCODE vM37 mouse GTF because it is a **comprehensive and high-quality reference annotation** containing well-curated gene structures for *Mus musculus* (mm10), ensuring accurate transcript assignment.

- **Assembly/Purpose:**

- The GTF annotation file contains the official gene and transcript coordinates for *Mus musculus* (mm10). By providing this to StringTie, the program can match our aligned reads to known genes and transcripts and calculate expression levels for each. This ensures our quantification is biologically meaningful and not just based on raw read positions.
- We created a script for the assembly and annotation which consist of:
 1. **Uses StringTie to process sorted BAM files from the alignment step** — these BAM files are the outputs from HISAT2, sorted by genomic coordinates so StringTie can efficiently reconstruct transcripts
 2. **Guides the assembly with the GTF annotation** — this is called reference-guided assembly, which ensures the reconstructed transcripts match known gene models from GENCODE vM37, while still allowing detection of novel transcripts if present.
- Run time: ~ 3 minutes per sample

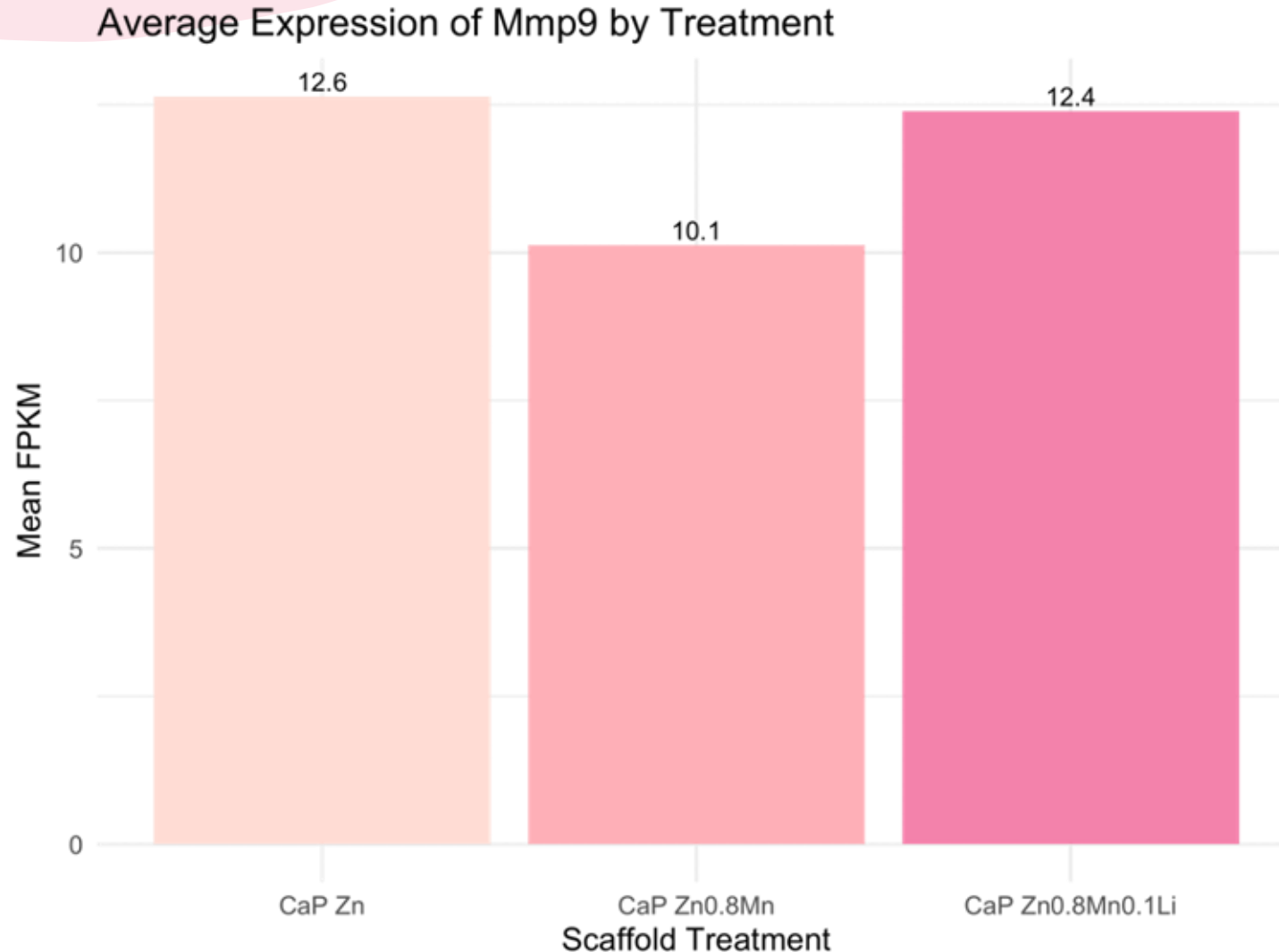
- **Output:**

- Transcript level annotation files (.gtf)
- Gene abundance tables (*_gene_abund.tab) with Gene IDs, names, and FPKM values
- We then use the gene abundance file for expression analysis of Mmp2 and Mmp9 in R

Gene Expression Analysis & Visualization

- Once in R, we:
 - Imported the *_gene_abund.tab files from StringTie output.
 - Filtered the data to extract rows corresponding to MMP2 and MMP9.
 - Created a metadata table assigning each sample to its treatment group (treatment info obtained from the NCBI dataset page).
 - Merged metadata with expression data.
 - Calculated the mean FPKM (Fragments Per Kilobase of transcript per Million mapped reads — a normalized measure of gene expression) for each treatment group and each gene.
 - Created visualizations:
 - Bar plot of average MMP2 and MMP9 expression by treatment.
 - Dot plot showing replicate-level variation in FPKM values.

BarPlot of MMP9



Result:

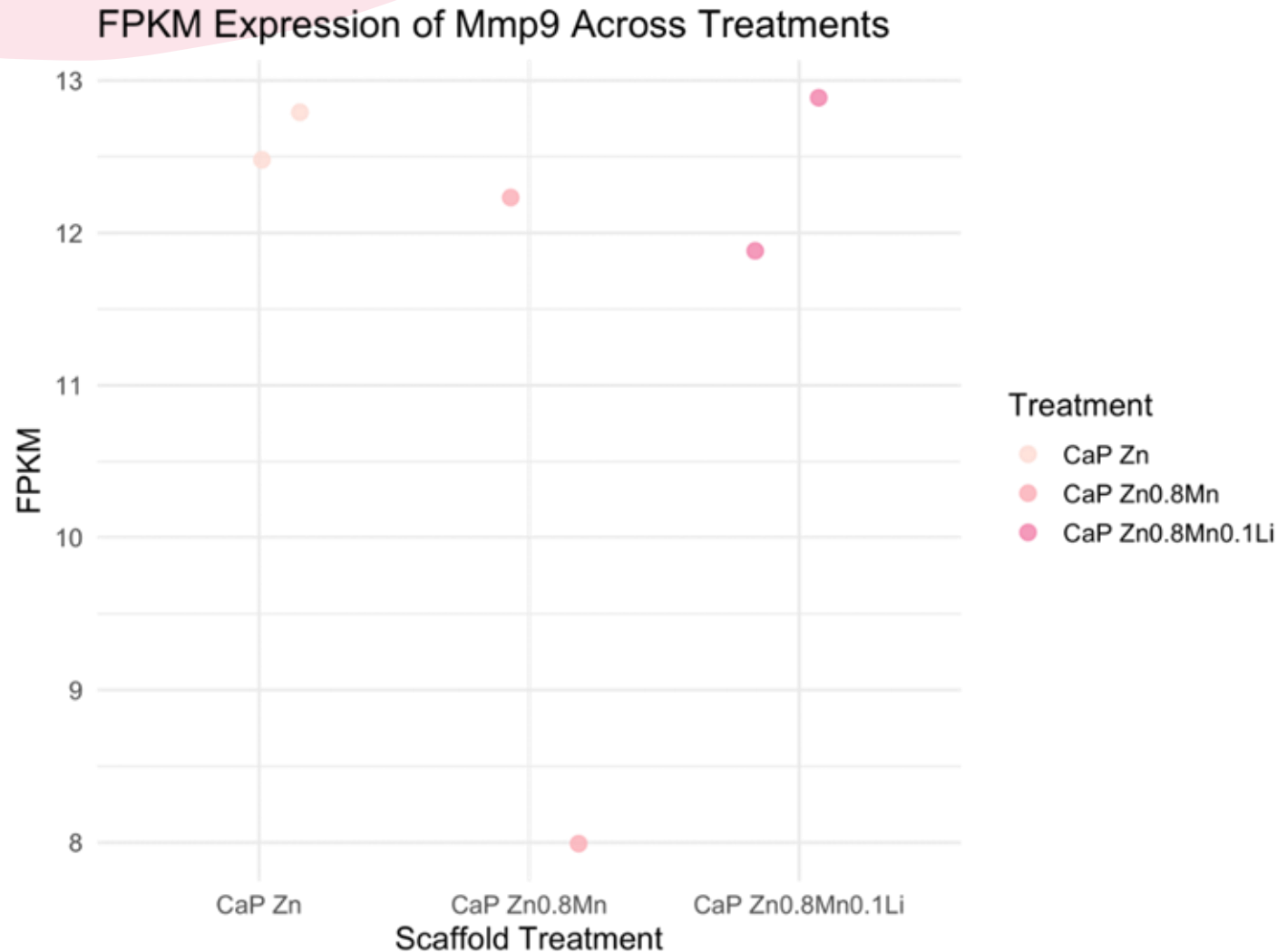
This plot shows the average FPKM values for MMP9 across three scaffold treatments.

- **CaP Zn** had the highest mean expression (~12.6), followed closely by **CaP Zn0.8Mn0.1Li** (~12.4).
- **CaP Zn0.8Mn** had the lowest mean expression (~10.1) compared to the other treatments.

Interpretation:

MMP9 is an enzyme involved in degradation and tissue remodeling. Cap Zn has a higher expression of MMP9 which suggests a stronger inflammatory and tissue remodeling response compared to the other treatments.

Dot plot of MMP9



Result:

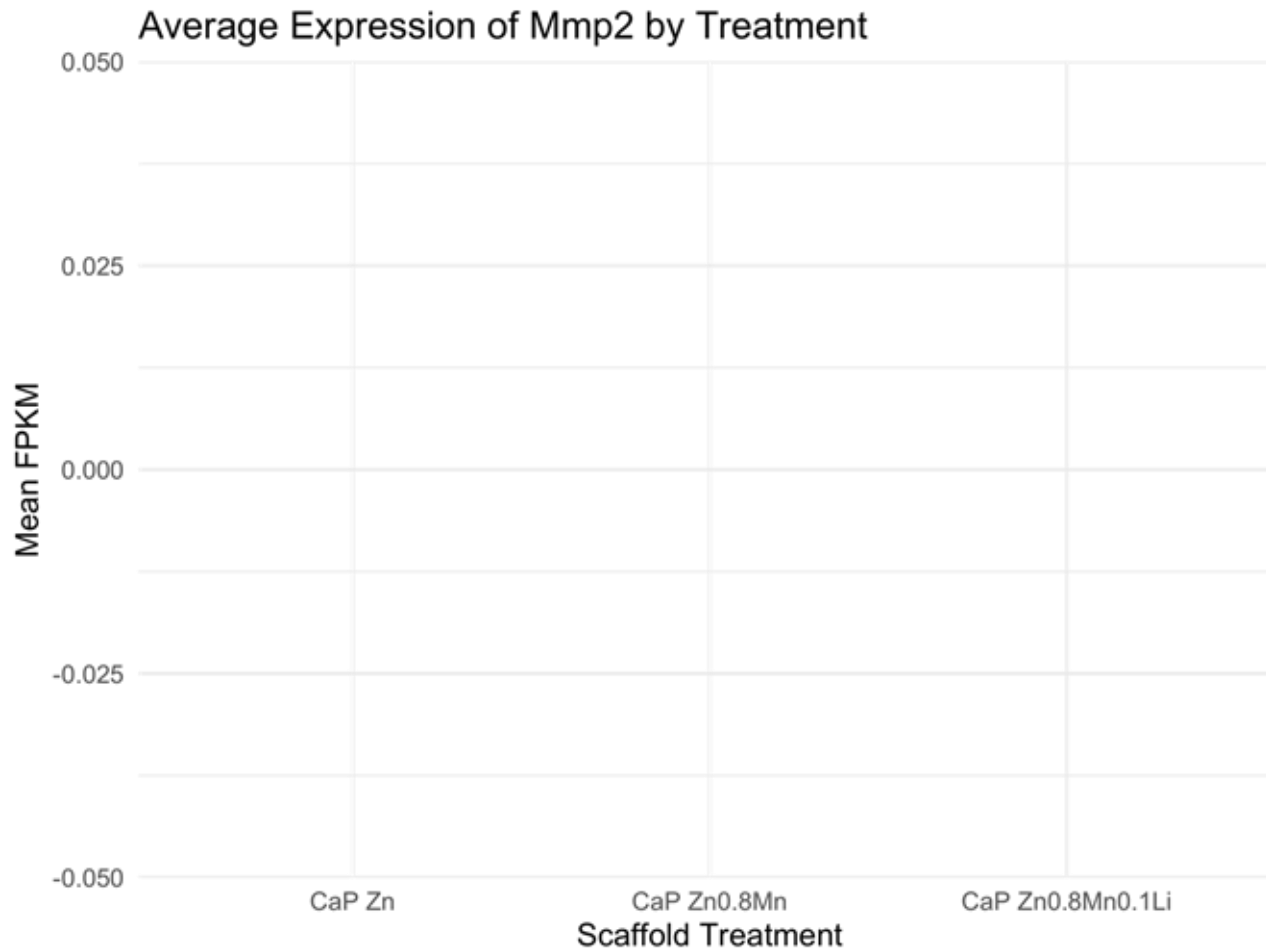
This dot plot shows the FPKM expression of MMP9 for each individual replicate within the three scaffold treatments.

- CaP Zn replicates generally had the highest values, followed closely by CaP Zn0.8Mn0.1Li.
- CaP Zn0.8Mn replicates showed lower expression, with one replicate being lower than the others.

Interpretation:

The plot highlights variability between replicates, especially in the CaP Zn0.8Mn group. Since MMP9 is associated with inflammation and tissue remodeling, the higher and more consistent expression in CaP Zn suggests it may induce a stronger macrophage response compared to the other treatments.

BarPlot of MMP2



Result:

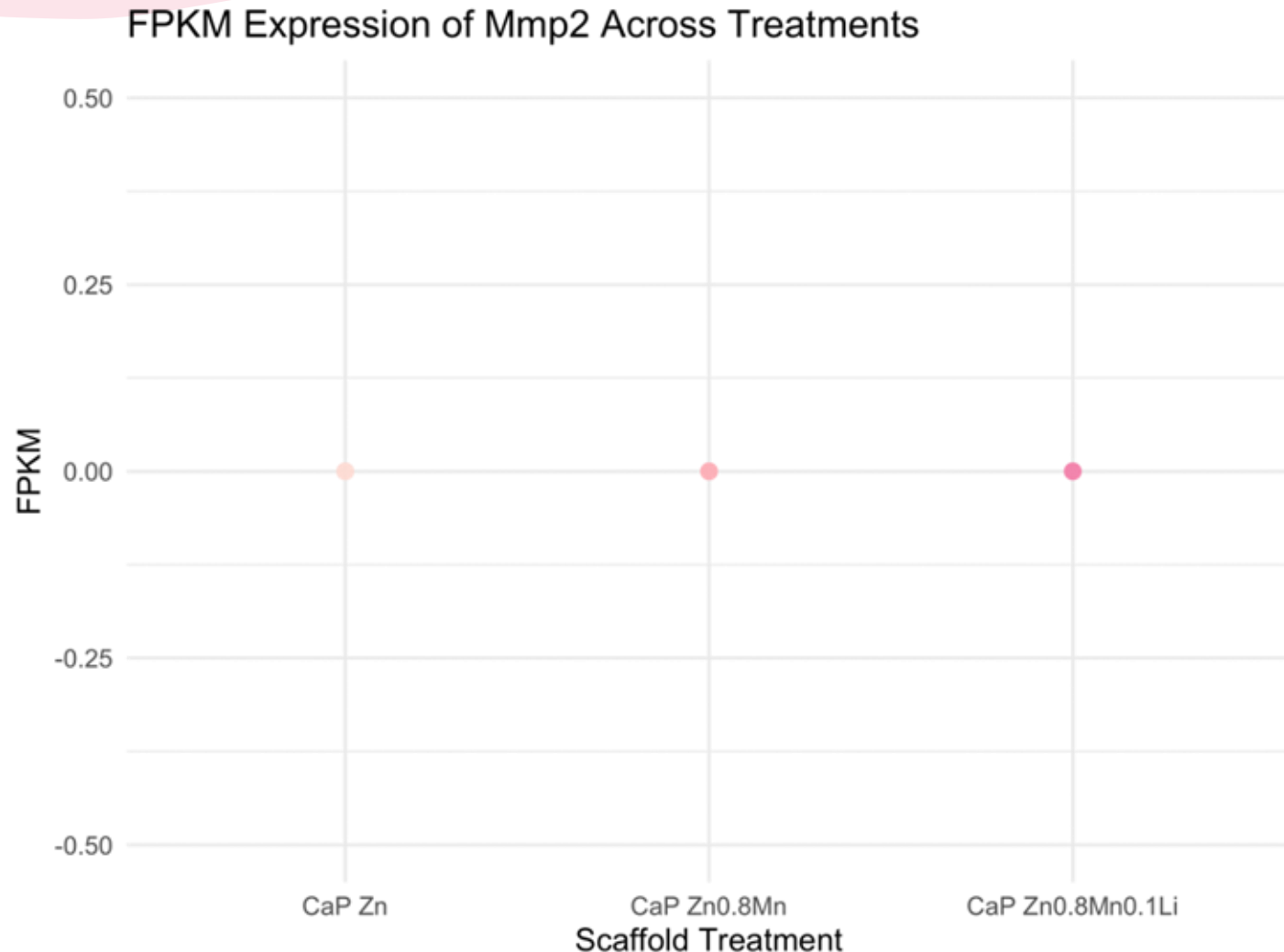
This plot shows the average MMP2 expression (measured as FPKM) for each scaffold treatment.

- All three groups — CaP Zn, CaP Zn0.8Mn, and CaP Zn0.8Mn0.1Li — had values at zero. This means MMP2 was barely expressed or not expressed at all in our samples.

Interpretation:

MMP2 is an enzyme that helps break down the extracellular matrix during tissue remodeling. Since its expression was extremely low in all treatments, the scaffolds in this experiment did not seem to trigger MMP2 activity in the macrophages.

Dot plot of MMP2



Result

- The dot plot shows the FPKM expression of MMP2 for each individual replicate within the three scaffold treatments.

Interpretation:

- All replicates show an expression value of 0, indicating that MMP2 was not detectably expressed in these macrophages under any scaffold condition.

Limitations/ Future Plan

- **Limitations**

- **Small sample size** – Only 2 biological replicates per scaffold type
- **Public dataset** – Limited control over experimental variables such as scaffold preparation and exposure time

- **Future plan**

- **Statistical Analysis** - Running a statistical test such as anova

Conclusion

In conclusion, our project used RNA-seq analysis to examine how macrophage *MMP2* and *MMP9* expression changes in response to zinc-based CaP scaffold treatments. We found that *MMP9* expression was highest in *CaP Zn* compared to other scaffolds, suggesting a stronger tissue remodeling or inflammatory response. In contrast, *MMP2* showed no detectable expression under any treatment, indicating it may not play a significant role in the macrophage response in this context.

Reference

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